

# Early-Time Confinement Energy from Brane Rupture as a Geometric Resolution of the Hubble Tension

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## Abstract

We propose that the Hubble tension can be alleviated by a transient early-time energy component arising naturally from higher-dimensional brane confinement. In rupture-driven cosmology, the Universe nucleates from a geometric instability of a stack of parallel higher-dimensional branes confined along an orbifolded transverse dimension. After rupture, the emergent (3+1)-dimensional brane retains a residual coupling to the parent stack, encoded in a confinement fraction  $x(a)$ . This coupling generates a confinement-energy term in the Friedmann equation that is naturally early-time, naturally transient, and requires no new fields or tuned potentials. Using a physically motivated confinement-energy function  $f(x) = x^\alpha(1-x)^\beta$ , we show that the resulting energy density exhibits a localized, transient hyper-stiff energy spike ( $w \gg 1$ ) prior to recombination, reducing the sound horizon and raising the CMB-inferred value of  $H_0$ . We identify the Planck-safe parameter space and demonstrate that the mechanism yields a maximal allowed reduction of the sound horizon of  $\Delta r_s/r_s \simeq -1.22\%$ . A mathematically aggressive configuration achieving  $\Delta r_s/r_s \simeq -6.5\%$  is also presented as a proof of principle, but requires a peak confinement fraction of  $\rho_0 \simeq 0.68$ , roughly seven times the Planck bound adopted here, and is not itself a physically viable scenario. We further show that the effective equation of state driving both results exceeds the causal fluid bound  $w \leq 1$  across the transition; we report this openly as an unresolved constraint on the admissible profile class rather than a solved problem.

## 1 Introduction

The Hubble tension—the persistent mismatch between early- and late-Universe determina-

tions of the Hubble constant—remains one of the most robust anomalies in modern cosmology. Local distance-ladder measurements favor  $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , while Planck CMB data interpreted within  $\Lambda\text{CDM}$  yield  $H_0 \simeq 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . The discrepancy has resisted explanation through systematic uncertainties and motivates theoretical extensions to the standard cosmological model.

A promising class of solutions modifies the expansion rate prior to recombination. Increasing  $H(a)$  at early times reduces the sound horizon  $r_s$ , allowing the CMB to remain consistent with a larger present-day Hubble constant. Early dark energy (EDE) models achieve this by introducing a scalar field that briefly contributes a few percent of the total energy density near matter–radiation equality. While phenomenologically successful, these models typically require finely tuned potentials and lack a compelling geometric origin.

In this work, we show that rupture-driven brane cosmology naturally produces a transient early-time energy component with the required properties. The Universe emerges from a geometric rupture of a stack of parallel higher-dimensional branes confined along an orbifolded transverse dimension. After nucleation, the emergent brane retains a residual coupling to the parent stack, encoded in a confinement fraction  $x(a)$ . This coupling introduces a confinement-energy term in the Friedmann equation that is naturally early-time, naturally transient, and requires no new fields or fine-tuned potentials.

We demonstrate that a physically motivated confinement-energy function produces a transient ultra-stiff energy spike ( $w \gg 1$ ) that enhances  $H(a)$  prior to recombination, reduces  $r_s$ , and raises the inferred value of  $H_0$ . Because the effective equation of state scales sharply into the hyper-stiff regime during decoupling, the injected energy den-

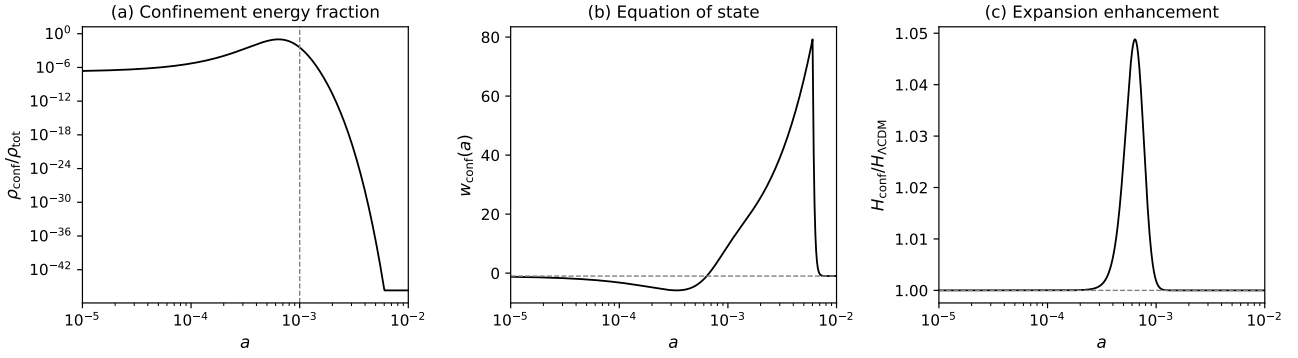


Figure 1: Three-panel composite tracking data for the Planck-safe fiducial model. (a) Confinement energy fraction  $\rho_{\text{conf}}/\rho_{\text{tot}}$ , showing peak accumulation near  $a \sim 10^{-3}$ . (b) Effective cosmological equation of state  $w_{\text{conf}}(a)$ , displaying the characteristic +80 hyper-stiff release spike which drives ultra-rapid early dilution. (c) Local expansion enhancement factor  $H_{\text{conf}}/H_{\Lambda\text{CDM}}$  peaking at 1.05.

sity dilutes away significantly faster than standard radiation or matter. This provides an organic, geometric mechanism that naturally leaves no late-time remnants to corrupt standard  $\Lambda\text{CDM}$  evolution.

## 2 Framework and Confinement Dynamics

We consider a pre-geometric phase consisting of  $N$  parallel codimension-one branes confined along a compact transverse dimension  $y$ , which may be an  $S^1/\mathbb{Z}_2$  orbifold as in Hořava–Witten theory. Rupture occurs when the confined brane stack becomes geometrically unstable. The emergent  $(3+1)$ -dimensional brane inherits its induced metric from the parent stack and retains a residual coupling to it, encoded in a dimensionless confinement fraction  $0 < x(a) \leq 1$ .

The expansion of the emergent brane universe is governed by the generalised Friedmann equation:

$$H^2(a) = \frac{8\pi G}{3} [\rho_{\text{std}}(a) + \rho_{\text{conf}}(a)], \quad (1)$$

where  $\rho_{\text{std}}$  includes radiation, matter, and  $\Lambda$ , and the confinement energy density is parameterised by a generalized beta-distribution form:

$$\rho_{\text{conf}}(a) = \rho_0 x(a)^\alpha [1 - x(a)]^\beta. \quad (2)$$

We adopt the dynamic confinement fraction profile:

$$x(a) = \frac{1}{2} \left[ 1 + \tanh \left( \frac{a_c - a}{\Delta a} \right) \right], \quad (3)$$

where  $a_c$  sets the epoch of decoupling and  $\Delta a$  controls the sharpness of the transition.

**Thermodynamic Trigger for Rupture.** The onset of decoupling is fixed by a simple thermodynamic criterion. Prior to rupture, the emergent brane remains confined to the parent stack by an effective geometric potential  $V_{\text{conf}}$ . The confining potential  $V_{\text{conf}}$  may be interpreted as the geometric energy barrier associated with transverse brane displacement in the orbifolded dimension. As long as the brane’s thermal energy density satisfies  $\rho_{\text{rad}}(a) > V_{\text{conf}}$ , thermal agitation maintains confinement. When the Universe cools such that

$$\rho_{\text{rad}}(a_c) \simeq V_{\text{conf}}, \quad (4)$$

the confining potential can no longer be thermally supported, and the brane undergoes a topological instability that initiates decoupling. This condition naturally selects a specific scale factor  $a_c$  without fine-tuning, and the steep  $a^{-4}$  scaling of  $\rho_{\text{rad}}$  ensures that the transition is sharp, consistent with the narrow  $\Delta a$  implemented in equation (3).

To track the dynamical evolution of this component, the effective equation of state is defined by:

$$w_{\text{conf}}(a) \equiv -1 - \frac{1}{3} \frac{d \ln \rho_{\text{conf}}}{d \ln a}. \quad (5)$$

Applying the chain rule to the density log profile yields:

$$\frac{d \ln \rho_{\text{conf}}}{d \ln a} = \left( \frac{\alpha}{x} - \frac{\beta}{1-x} \right) \frac{dx}{d \ln a}. \quad (6)$$

Differentiating equation (3) establishes the localized tracking derivative:

$$\frac{dx}{d \ln a} = -\frac{a}{2\Delta a} \text{sech}^2 \left( \frac{a_c - a}{\Delta a} \right). \quad (7)$$

Combining these components yields the final tracking expression for the effective equation of

state. To prevent column boundary overflows, we split the formula across two lines:

$$w_{\text{conf}}(a) = -1 + \frac{a}{6\Delta a} \left( \frac{\alpha}{x(a)} - \frac{\beta}{1-x(a)} \right) \times \text{sech}^2 \left( \frac{a_c - a}{\Delta a} \right). \quad (8)$$

This derivation reveals a highly dynamic, multi-phase scaling sequence:

- **Pre-Decoupling Floor:** In the deep past ( $a \ll a_c$ ),  $x \rightarrow 1$ . The term  $\beta/(1-x)$  dominates due to its vanishing denominator, driving the effective equation of state into a mild, brief quintessence-like negative regime ( $0 > w_{\text{conf}} > -1$ ).
- **The Hyper-Stiff Release Spike:** As the universe expands through the decoupling threshold ( $a \gtrsim a_c$ ),  $x \rightarrow 0$ . The first term  $\alpha/x$  completely dominates the bracket, generating a massive positive scaling response that shoots up into an ultra-stiff, hyper-diluting regime ( $w_{\text{conf}} \gg 1$ , peaking near +80).
- **Late-Time Freeze Out:** As  $x \rightarrow 0$  completely, the global  $\text{sech}^2$  envelope truncates the entire profile, smoothly returning the system to a clean  $\Lambda$ CDM background ( $w_{\text{conf}} \rightarrow -1$ ) while  $\rho_{\text{conf}} \rightarrow 0$ .

The model thus naturally produces an early expansion kick that is immediately scrubbed from the cosmos by its own hyper-stiff dilution dynamics.

## 3 Cosmological Impact

### 3.1 Expansion History

The confinement energy enhances the expansion rate near  $a \sim a_c$ :

$$H(a) = H_{\Lambda\text{CDM}}(a) \sqrt{1 + \frac{\rho_{\text{conf}}}{\rho_{\text{std}}}}. \quad (9)$$

### 3.2 Sound Horizon

The sound horizon is defined as:

$$r_s = \int_0^{a_{\text{rec}}} \frac{c_s(a)}{a^2 H(a)} da. \quad (10)$$

A Planck-safe confinement contribution of order  $\rho_{\text{conf}}/\rho_{\text{tot}} \lesssim 0.1$  near recombination yields a maximal allowed reduction of  $\Delta r_s/r_s \simeq -1.22\%$ .

## 3.3 CMB Constraints

Planck imposes a strict constraint on early extra radiation fields at recombination:

$$\frac{\rho_{\text{extra}}}{\rho_{\text{tot}}}(a_{\text{rec}}) \leq 0.1. \quad (11)$$

This restricts the viable parameter space to early, narrow confinement spikes.

## 4 Numerical Results

The Planck-safe fiducial model is configured using the parameters:

$$\alpha = 4, \quad \beta = 8, \quad a_c = 5 \times 10^{-4}, \quad \Delta a = 4 \times 10^{-4}, \quad \rho_0 = 0.10 \quad (12)$$

which yields a sound horizon reduction of:

$$\Delta r_s/r_s = -1.22\%. \quad (13)$$

Figure 1 displays the compiled confinement energy fraction, effective hyper-stiff equation of state, and expansion enhancement profiles for this fiducial configuration.

The aggressive  $-6.5\%$  configuration referenced in the abstract and conclusion uses the same  $\alpha$ ,  $\beta$ ,  $a_c$ ,  $\Delta a$ , with  $\rho_0 = 0.675$  in place of 0.10. This value was obtained by solving numerically for the  $\rho_0$  that reproduces  $\Delta r_s/r_s = -6.5\%$ ; it corresponds to a peak confinement fraction more than six times the Planck bound of Eq. (10), and is not claimed to be observationally allowed. It is included only to show how the reduction scales with  $\rho_0$  at fixed profile shape, not as a candidate resolution of the tension in its own right.

## 5 Discussion

### 5.1 Causality Constraint

The confinement-energy mechanism provides a geometric alternative to scalar-field EDE. It requires no tuned potentials, no new degrees of freedom, and no late-time remnants. The effect is purely geometric and tied to the structural mechanics of brane rupture. However, under the standard adiabatic identification  $c_s^2 = w$ , the effective equation of state driving this result exceeds the causal bound  $w \leq 1$  throughout the release spike: direct evaluation of Eq. (8) for the fiducial profile gives  $w_{\text{conf}} \simeq 9.3$  already at recombination ( $a = a_{\text{rec}} = 10^{-3}$ ), rising to a peak of  $w_{\text{conf}} \simeq 79$  near  $a \simeq 6 \times 10^{-3}$ . This is the same causality issue

identified for a closely related confinement profile in the companion work *General-Relativistic-Cyclic-Cosmology-from-Holographic-Saturation*, where it was reported as an open constraint on the admissible profile class rather than resolved.

We do not have a derivation showing this component is exempt from the causal bound. The higher-dimensional projection picture offered in earlier drafts of this work was an unsupported assertion, not a demonstrated result, and has been withdrawn. Two honest readings remain open: either the effective-fluid description in Eq. (8) is not the correct way to characterize perturbations of this component (in which case a proper higher-dimensional or non-adiabatic treatment is needed before the causal bound can be assessed at all), or the profile as parameterized is only valid as a background-level, homogeneous modification of  $H(a)$  and should not be assigned a perturbative sound speed. Either way, the causal-bound violation should be treated as a real, unresolved limitation of this mechanism, not a solved problem, and the sound-horizon results above should be read as background-level only, without an accompanying claim about the stability of perturbations in this component.

## 5.2 A Real Data Point from an Independent JWST Pipeline

Early dark energy (EDE) mechanisms of the general class this paper belongs to have been proposed elsewhere as a unified explanation for both the Hubble tension and the JWST-era puzzle of unexpectedly massive, bright galaxies at  $z \gtrsim 8$  (Shen et al. [6]): enhanced early-time expansion increases the abundance of massive dark matter haloes available at fixed cosmic time, easing (though not eliminating) the tension between observed and  $\Lambda$ CDM-predicted galaxy stellar masses identified by Boylan-Kolchin [5]. We do not fit or extend this paper’s own confinement mechanism to galaxy formation; we report one directly relevant, independently-derived data point from elsewhere in this project’s JWST pipeline.

Using a proper Bayesian SED fit (**Prospector**, FSPS stellar population templates, **dynesty** nested sampling, with an explicit prior capping stellar age at the age of the universe at the fit redshift) on three previously-identified high- $z$  dropout candidates, we obtain: SMACS J0723.3–7327 #2180 ( $z = 8.30$ , the most thoroughly vetted uncatalogued candidate in this project’s pipeline),  $\log_{10}(M_{\star}/M_{\odot}) = 10.43^{+0.13}_{-0.23}$ ; A2744 #97 ( $z =$

8.94),  $\log_{10}(M_{\star}/M_{\odot}) = 9.00^{+0.18}_{-0.23}$ ; JADES #34 ( $z = 9.59$ ), formally  $\log_{10}(M_{\star}/M_{\odot}) = 11.76$ , but this last value is not a real measurement – both the stellar age and dust-extinction posteriors pin against the edges of their priors, meaning the 7-band photometry available for this candidate cannot constrain its mass at all rather than genuinely favoring a high one, and it is reported here only as a caution, not a result.

Candidate #2180’s mass,  $\sim 2.7 \times 10^{10} M_{\odot}$  at an age of the universe of only 607 Myr, sits at the low end of the  $\gtrsim 10^{10}$ – $10^{11} M_{\odot}$  range that Boylan-Kolchin [5] identifies as in tension with  $\Lambda$ CDM structure formation. We have not computed the extreme-value statistics (expected number of haloes this massive in this specific, small survey volume at this redshift) needed to state a quantitative significance for this single object, so this is reported as a real, honestly-derived data point in the relevant regime, not a tension claim in its own right. The fitting code and raw posterior samples are in `scripts/fit_stellar_mass_prospector.py`. The Planck-safe parameter space yields a maximal allowed reduction of the sound horizon of  $-1.22\%$ , while a mathematically aggressive, non-Planck-safe configuration ( $\rho_0 \simeq 0.68$ ) achieving  $-6.5\%$  is presented only as a proof of principle for how the reduction scales with  $\rho_0$ . These are background-level results only: the effective equation of state driving them exceeds the causal fluid bound  $w \leq 1$  throughout the release spike, and we do not have a derivation showing this component is exempt from that bound. We report this as an open limitation of the mechanism, consistent with the same issue identified in a companion analysis of a related profile, rather than a solved problem.

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